Ansys Fluent Simulation Report

Analyst	hp
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System Information

Application	Fluent
Settings	3d, double precision, pressure-based, inviscid
Version	24.1.0-10184
Source Revision	5b3f9fb3c8
Build Time	Nov 22 2023 10:32:41 EST
CPU	AMD Ryzen 5 5625U with Radeon Graphics
OS	Windows

Geometry and Mesh

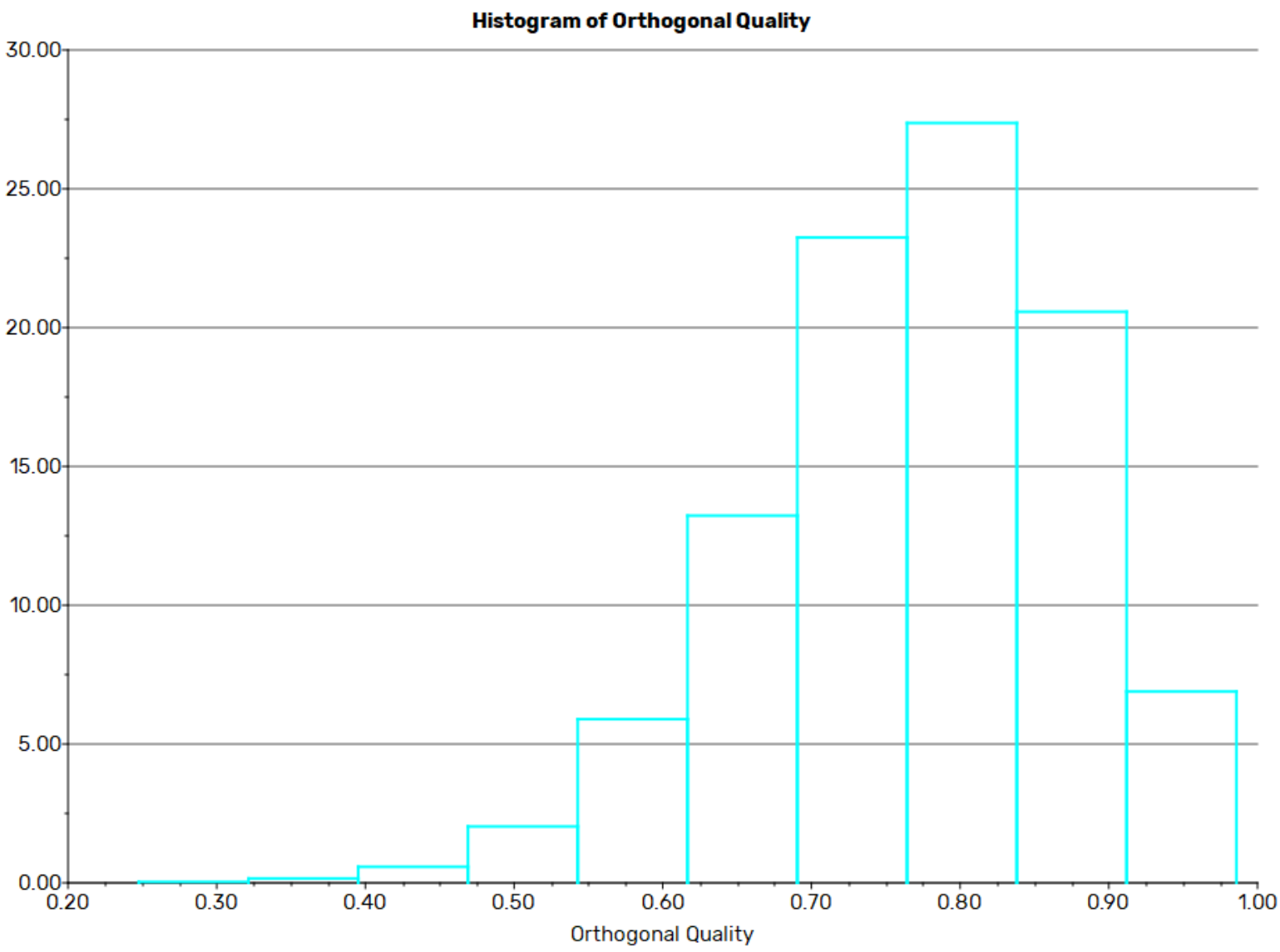
Mesh Size

Cells	Faces	Nodes
16658	34604	3465

Mesh Quality

Name	Type	Min Orthogonal Quality	Max Aspect Ratio
part2	Tet Cell	0.24751976	12.133476

Orthogonal Quality



Simulation Setup

Physics

Models

Model	Settings
Space	3D
Time	Steady
Viscous	Inviscid
Heat Transfer	Enabled

Material Properties

▬ Fluid	
▬ rocketfluid	
Density	ideal gas
Cp (Specific Heat)	2494 J/(kg K)
Molecular Weight	20 kg/kmol
▬ Solid	
aluminum	

Cell Zone Conditions

▬ Fluid	
▬ part2	
Material Name	rocketfluid
Specify source terms?	no
Specify fixed values?	no
Frame Motion?	no
Porous zone?	no
3D Fan Zone?	no

Boundary Conditions

▬ Inlet	
▬ inlet	
Reference Frame	Absolute
Gauge Total Pressure [Pa]	9800000
Supersonic/Initial Gauge Pressure [Pa]	9800000
Total Temperature [K]	3710
Direction Specification Method	Normal to Boundary
Build artificial walls to prevent reverse flow?	no
▬ Outlet	

▬ outlet	
Backflow Reference Frame	Absolute
Gauge Pressure [Pa]	101325
Pressure Profile Multiplier	1
Backflow Total Temperature [K]	300
Backflow Direction Specification Method	Normal to Boundary
Backflow Pressure Specification	Total Pressure
Build artificial walls to prevent reverse flow?	no
Radial Equilibrium Pressure Distribution	no
Average Pressure Specification?	no
Specify targeted mass flow rate	no
▬ Wall	
wall	wall

Reference Values

Area	1 m^2
Density	1.225 kg/m^3
Enthalpy	0 J/kg
Length	1 m
Pressure	0 Pa
Temperature	288.16 K
Velocity	1 m/s
Ratio of Specific Heats	1.4
Yplus for Heat Tran. Coef.	300
Reference Zone	part2

Solver Settings

▬ Equations	
Flow	True
Energy	True
▬ Numerics	
Absolute Velocity Formulation	True
▬ Pseudo Time Explicit Relaxation Factors	
Density	1
Body Forces	1
Energy	0.75
Temperature	0.75
Explicit Momentum	0.5
Explicit Pressure	0.5
▬ Pressure-Velocity Coupling	
Type	Coupled
Pseudo Time Method (Global Time Step)	True
▬ Discretization Scheme	

Pressure	Second Order
Density	Second Order Upwind
Momentum	Second Order Upwind
Energy	Second Order Upwind
− Solution Limits	
Minimum Absolute Pressure [Pa]	1
Maximum Absolute Pressure [Pa]	5e+10
Minimum Static Temperature [K]	1
Maximum Static Temperature [K]	5000

Run Information

Number of Machines	1
Number of Cores	1
Case Read	8.072 seconds
Iteration	13.522 seconds
AMG	7.001 seconds
Virtual Current Memory	0.448498 GB
Virtual Peak Memory	0.632511 GB
Memory Per M Cell	8.02351

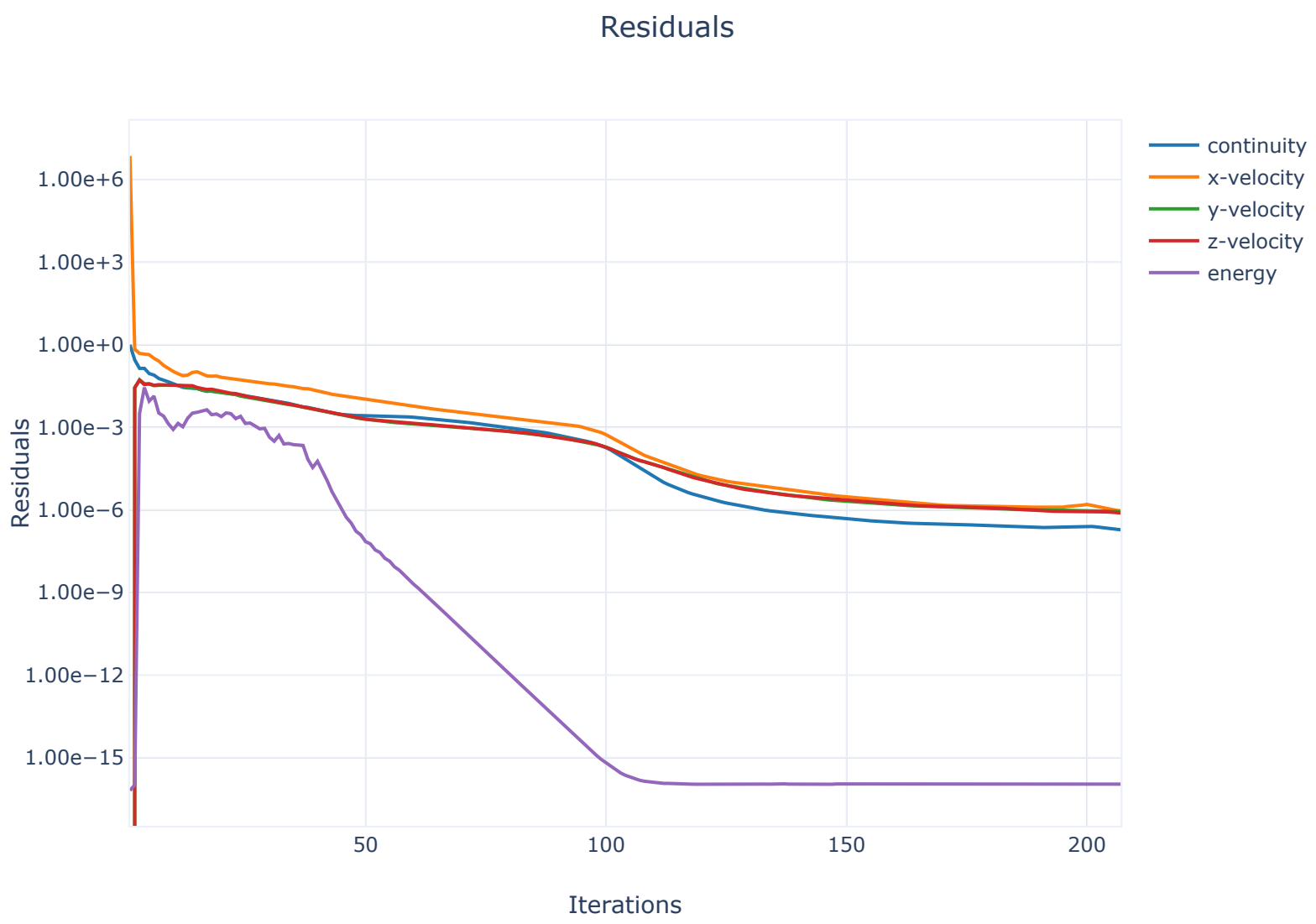
Solution Status

Iterations: 207

	Value	Absolute Criteria	Convergence Status
continuity	1.911693e-07	1e-06	Converged
x-velocity	9.308841e-07	1e-06	Converged
y-velocity	8.601627e-07	1e-06	Converged
z-velocity	7.816464e-07	1e-06	Converged
energy	1.099769e-16	1e-06	Converged

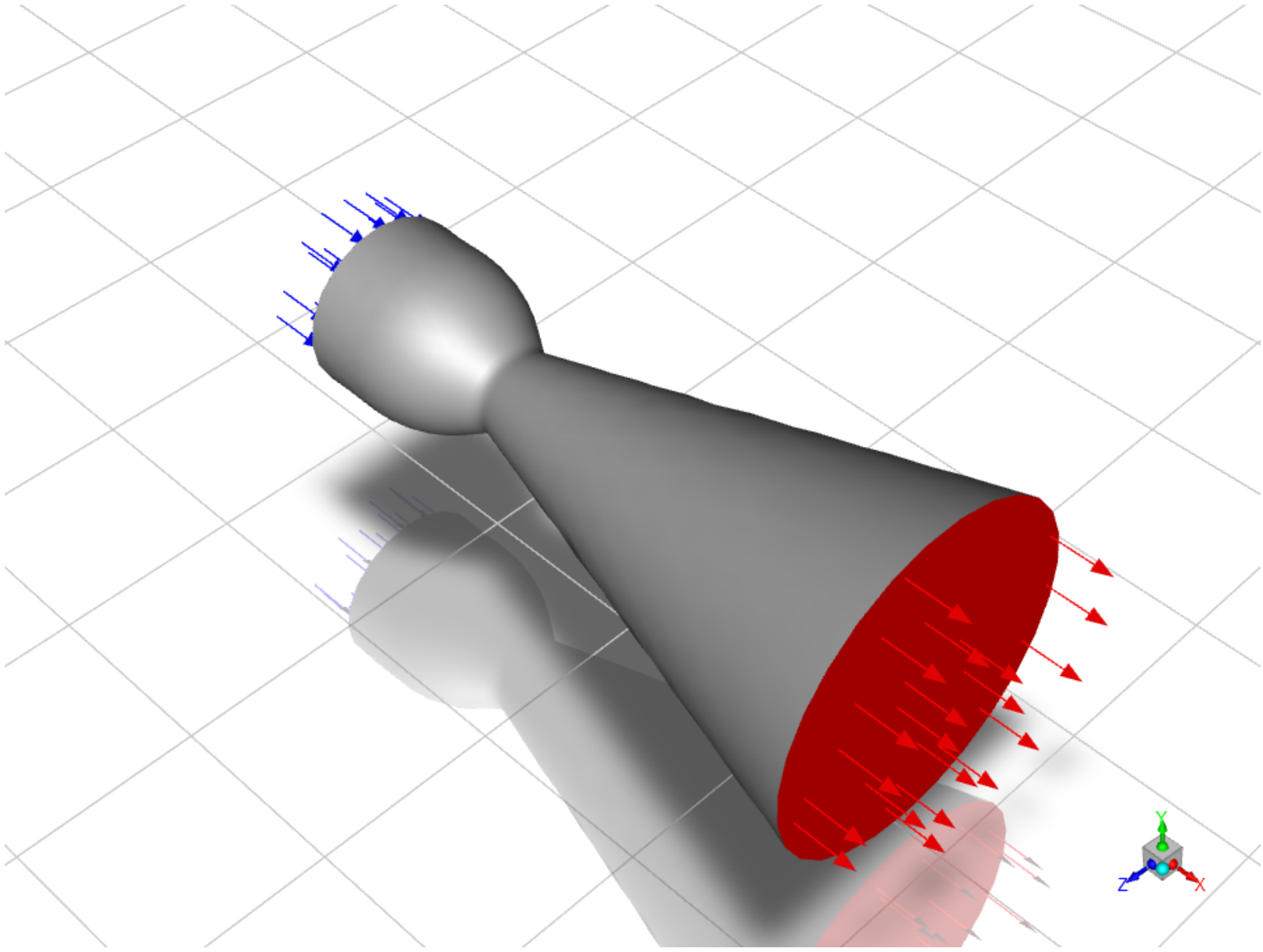
Plots

Residuals



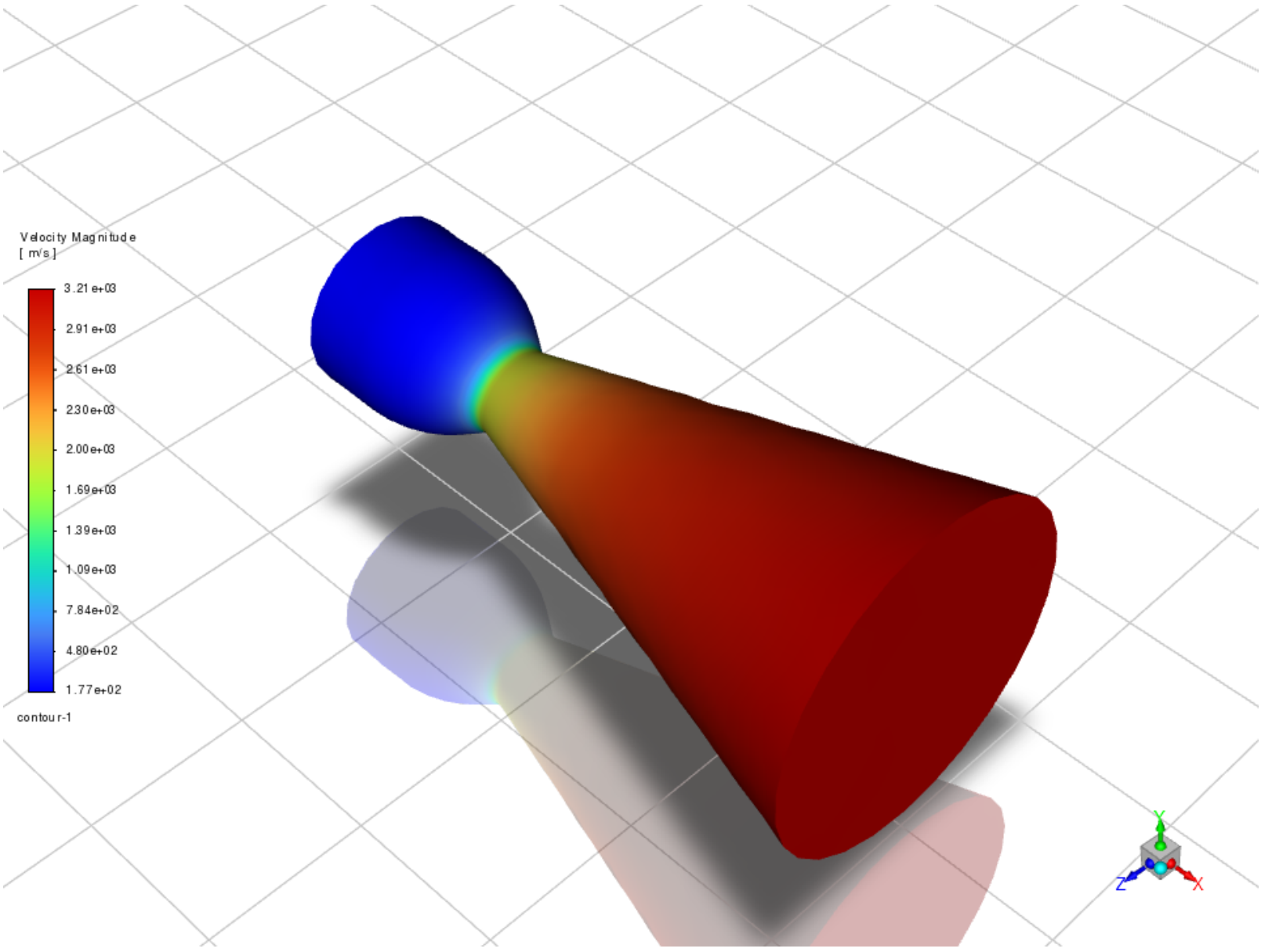
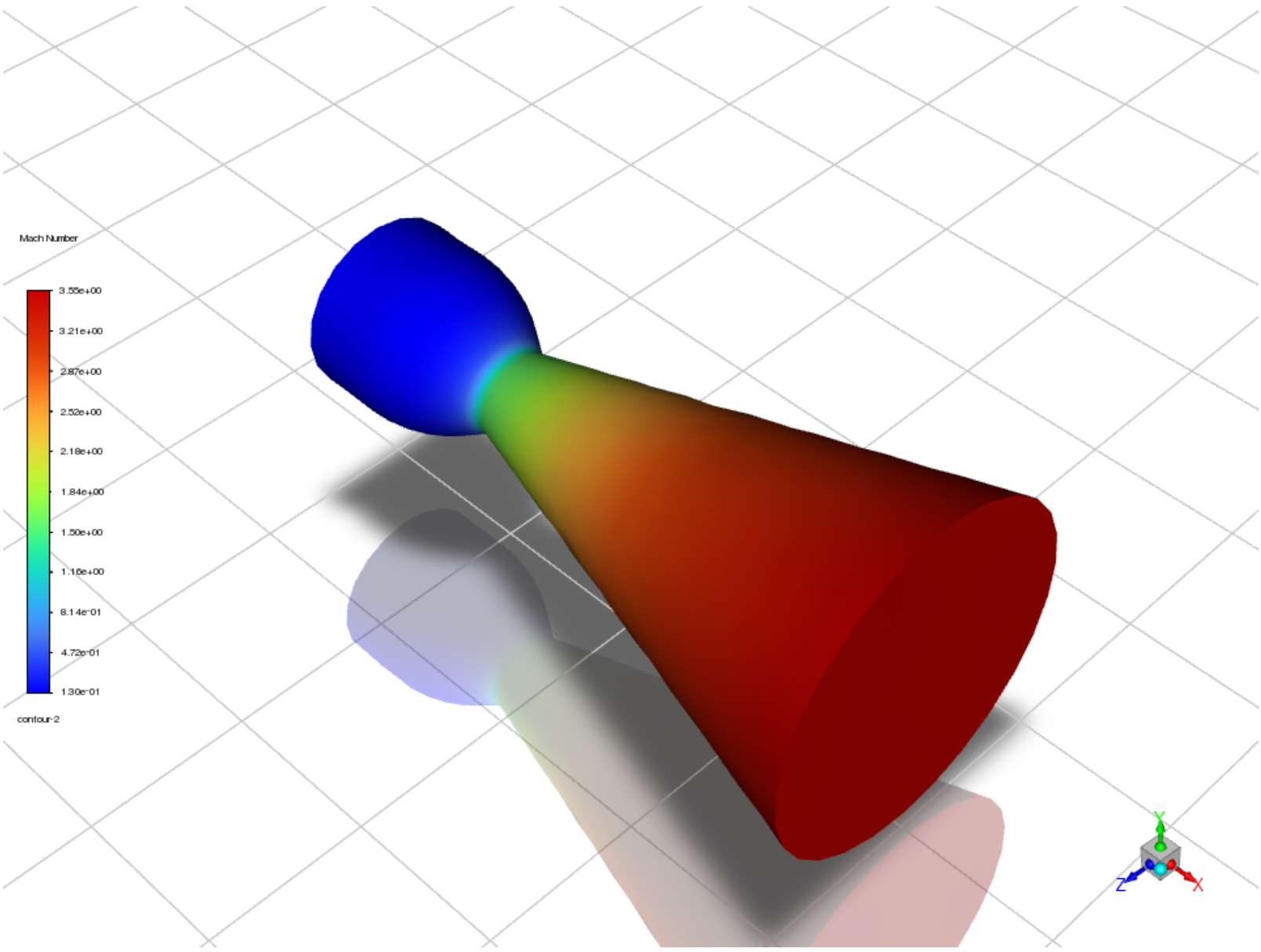
Mesh

mesh-1



Contours

contour-1



contour-2